

Load Interconnection Combined Feasibility / System Impact Study Report

For Queue Project Q15-0542

Debary 138 kV
December 20, 2016

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q15-0542. The Interconnection Customer proposes a 187.0 MW large load (data center) facility interconnecting at the Debary 138 kV substation in the Pennsylvania area. The request was received on November 25, 2015. The study evaluated system intact and single contingency conditions using the regional planning case, including 284 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 1 overloaded facility. The total estimated cost of the network upgrades allocated to this project is \$299,000.

3. Network Impacts

3.1 Load Deliverability Analysis

The E CALAMUS - DEBARY 138 KV Line is overloaded to 111.1% of its 679 MVA emergency rating for the loss of the STATE CAMPUS - AMERICAN RIV 138 KV. The project contributes 49.0% DFAX (91.7 MW) to this overload.

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
E CALAMUS - DEBARY 138 KV Line	Rebuild line with higher-capacity conductor	\$2,485,000	12% (\$299,000)
	Total allocated to project		\$299,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.